

Problem 8.4

Let i be a compound interest with equivalent compound discount d . Show the following:

(a) $\frac{1}{d} - \frac{1}{i} = 1$.

(b) $d \left(1 + \frac{i}{2} \right) = i \left(1 - \frac{d}{2} \right)$.

(c) $i\sqrt{1-d} = d\sqrt{1+i}$.

Hint: Use $i - d = id$ for all three identities.

Solution

Since i and d are equivalent effective rates, we use the standard relation

$$i - d = id.$$

(a)

Starting from the left-hand side,

$$\frac{1}{d} - \frac{1}{i} = \frac{i - d}{id}.$$

Using $i - d = id$,

$$\frac{i - d}{id} = \frac{id}{id} = 1.$$

Hence,

$$\boxed{\frac{1}{d} - \frac{1}{i} = 1}.$$

(b)

Consider the right-hand side:

$$i \left(1 - \frac{d}{2} \right) = i - \frac{id}{2}.$$

Since $id = i - d$, this becomes

$$i - \frac{i - d}{2} = \frac{2i - (i - d)}{2} = \frac{i + d}{2}.$$

Now consider the left-hand side:

$$d \left(1 + \frac{i}{2} \right) = d + \frac{id}{2}.$$

Again using $id = i - d$,

$$d + \frac{i - d}{2} = \frac{2d + i - d}{2} = \frac{i + d}{2}.$$

Since both sides equal $\frac{i+d}{2}$,

$$\boxed{d\left(1 + \frac{i}{2}\right) = i\left(1 - \frac{d}{2}\right)}.$$

(c)

Starting from the left-hand side,

$$d\sqrt{1+i} = \sqrt{d^2(1+i)}.$$

Expand inside the square root:

$$d^2(1+i) = d^2 + d^2i = d^2 + d(id).$$

Since $id = i - d$,

$$d^2 + d(i - d) = d^2 + di - d^2 = di.$$

Thus,

$$d\sqrt{1+i} = \sqrt{di}.$$

Now use $d = i - id = i(1 - d)$, so

$$\sqrt{di} = \sqrt{i^2(1-d)} = i\sqrt{1-d}.$$

Therefore,

$$\boxed{i\sqrt{1-d} = d\sqrt{1+i}}.$$